

Scalability & Future Plan

Date	3 JULY 2026
Team ID	SWTID-2026-2662
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	1 Mark

Scalability & Future Plan:

This document outlines how the current project solution can be scaled to handle larger user bases, increased data loads, or extended features in the future. It also captures the team's roadmap for enhancing and evolving the project beyond its current state.

Current System Limitations:

S.No	Limitation	Impact	Priority to Address (High / Medium / Low)
1	The platform currently supports only basic AI-powered settlement prediction and negotiation based on predefined financial inputs. It does not use real-time banking or credit bureau data.	Settlement recommendations may not fully reflect the user's latest financial status, affecting prediction accuracy.	High
2	The application uses SQLite as the database, which is suitable for development and small-scale deployments but not for handling large numbers of concurrent users.	Limited scalability and performance for enterprise-level or production environments.	Medium
3	The system currently relies on manual entry of loan and financial information and does not integrate directly with banks or financial institutions.	Users must manually update loan details, which may lead to outdated information and reduce automation.	High

Scalability Plan:

S.No	Scalability Aspect	Current State	Proposed Upgrade / Solution
1	User Load	The application supports a limited number of concurrent users using FastAPI with a SQLite database, suitable for	Deploy the application on cloud platforms (AWS, Azure, or Google Cloud), use load balancing, and replace SQLite with PostgreSQL/MySQL to support large numbers of concurrent users.

		development and small-scale deployment.	
2	Data Storage	User, loan, settlement, and AI history data are stored in a local SQLite database using SQLAlchemy ORM.	Migrate to PostgreSQL or MySQL with automated backup, replication, and cloud storage for improved scalability and reliability.
3	Performance	AI processing and financial calculations execute sequentially, and response time depends on Google Gemini API and backend processing.	Optimize FastAPI using asynchronous APIs, implement caching, background task processing, database indexing, and API performance optimization to reduce response time.
4	Security	The system implements JWT-based authentication and secure API access for users.	Enhance security by adding Multi-Factor Authentication (MFA), role-based access control (RBAC), HTTPS encryption, secure cloud deployment, API rate limiting, and continuous security monitoring.

Future Roadmap:

Phase	Planned Feature / Enhancement	Target Timeline	Expected Impact
Phase 2	Integrate real-time banking APIs and credit bureau data for automatic loan synchronization and financial analysis.	3–6 Months	Improves accuracy of financial analysis, reduces manual data entry, and enhances settlement recommendations.
Phase 3	Upgrade the database from SQLite to PostgreSQL/MySQL and deploy the application on cloud platforms (AWS, Azure, or Google Cloud).	6–9 Months	Enhances scalability, reliability, and supports a larger number of concurrent users.
Phase 4	Introduce advanced AI capabilities such as personalized financial planning, multilingual negotiation support, voice-enabled chatbot, and predictive financial analytics.	9–12 Months	Provides a more intelligent and personalized user experience while improving borrower engagement and decision-making.